Documentation chemical data input

This document contains the documentation for the chemical data input in the code **REMIX**. There are maximum 8 files to be read. The minimum of input files is 6 (the ones that end in *"_quim_loc.dat", "_sist_quim.dat"*, *"_tar_wat.dat", "_out_opts.dat", "CV_params.dat"* and *"_comp_opts.dat"*). The other two (ending in *"_tar_sol.dat"* and *"_tar_gas.dat"*) are optional, depending if the chemical system contains solids and/or gases. All of the files must contain the **same root** in their names, and they must live in the **same directory**.

We describe the input files with an example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory *"..\examples\calcite_eq"*.

Chemical system file: *"_sist_quim.dat"*

This file contains the information related to the chemical system (species, reactions, phases). It is read by the **Chemical System** class.

- **Line 1**: Title of the problem
- **Line 2**: Section separation
- **Line 3**: Section name
  - If Section name = 'PRIMARY AQUEOUS SPECIES' or 'AQUEOUS COMPLEXES':
    i. **Line 3a**: *sp_name* (string)
       i. Name of species in database ***"master25_modif.dat"***
    ii. **Line 3b**: end of section
  - If Section name = 'MINERALS' or 'GASES':
    i. **Line 3a**: *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
       i. *sp_name:* Name of species in database ***"master25_modif.dat"***
       ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
       iii. *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
    ii. **Line 3b**: end of section
  - If Section name = 'REDOX REACTIONS':
    i. **Line 3a**: *react_name* (string), *flag_eq* (logical)
       i. *react_name:* Name of redox reaction in database ***"Monod_DB.dat"***
       ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
    ii. **Line 3b**: end of section
  - If Section name = 'SURFACE COMPLEXES':
    i. **Line 3a**: *sp_name* (string)
       i. *sp_name:* Name of species in database ***"master25_modif.dat"***
    ii. **Line 3b**: end of section
- **Line 4**: Section separation
- **Line 5**: end of file

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-----------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'PRIMARY AQUEOUS SPECIES'
'h2o(p)'                                        ! name of primary aqueous species
'h+'
'ca+2'
'o2(aq)'
'hco3-'
'*'
'MINERALS'
'calcite' .true. .true.                         ! name of mineral, flag equilibrium, flag constant activity
'*' .true. .true.
'GASES'
'o2(g)'  .true. .true.                          ! name of gas, flag equilibrium, flag constant activity
'co2(g)'  .true. .true.                         ! name of gas, flag equilibrium, flag constant activity
'*'  .true. .true.
'-----------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'end'
```

Local chemistry file: *"_quim_loc.dat"*

This file contains the **local** chemical information (water types, mineral zones, gas zones, surfaces). It is read by the **Chemistry** class.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'INITIAL AND BOUNDARY WATER TYPES'
    - **Line 1**: *act_coeffs_model* (integer)
        0. Ideal
        1. Debye-Hückel
        2. Debye-Hückel extended
        3. Davies
        4. Truesdell-Jones
    - **Line 2**: *num_wat_types* (integer)
        i. Number of water types (includes initial, recharge, boundary, etc)
    - **Line 3**: *ind_wat_type* (integer), *temp* (real)
        i. *ind_wat_type*: index of water type
        ii. *temp*: temperature of solution (in Celsius)
    - **Line 3a**: *name* (string)
        i. Name of water type
    - **Line 3b**: water type data begins
        i. <u>*icon*</u>: initial condition type
            1. prescribed primary concentration
            2. prescribed total concentration
            3. prescribed activity
            4. primary species is in equilibrium with a constrain (mineral, gas or aqueous complex)
        ii. <u>*guess*</u>: Initial guess for primary species concentration
        iii. <u>*ctot*</u>: (all concentrations are expressed in molalities)
            i. *icon=1*: primary species concentration
            ii. *icon=2*: total concentration of primary species
            iii. *icon=3*: activity of primary species
            iv. *icon=4*: activity of constrain that is in equilibrium with this primary species
        iv. <u>*constrain*</u>: if *icon=4*, it is the species in equilibrium with the primary species
    - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
    - **Line 3d**: end of water type data
    - **Line 4**: end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'INITIAL AND BOUNDARY WATER TYPES'
0                                   ! activity coefficients model
2                                   ! number of water types
1   25.0                            ! index water type, temperature (C)
'boundary'                          ! name of water type
'    icon    guess    ctot    constrain'
'h2o(p)'   3    1d0    1d0       ''
'ca+2'     1    1d-9   1d-9      ''
'o2(aq)'   4    1d-9   2d-1      'o2(g)'
'hco3-'    4    1d-9   3.5d-4    'co2(g)'
'h+'       1    1d-7   1d-7      ''
'*'   0   0.0   0.0   ''
2   25.0                            ! index water type, temperature (C)
'resident'                          ! name of water type
'    icon    guess    ctot    constrain'
'h2o(p)'   3    1d0    1d0       ''
'ca+2'     1    1d-3   1d-3      ''
'o2(aq)'   1    1d-6   1d-6      ''
'hco3-'    4    1d-9   1d0       'calcite'
'h+'       1    1d-5   1d-5      ''
'*'   0   0.0   0.0   ''
'------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
```

- Section name: ‘INITIAL MINERAL ZONES’
    - **Line 1**: *num_min_zones* (integer)
        i. Number of mineral zones
    - **Line 2a**: *ind_min_zone* (integer), *temp* (real)
        i. *ind_min_zone*: index of mineral zone
        ii. *temp*: temperature of mineral zone (in Celsius)
    - **Line 2b**: *name* (string)
        i. Name of mineral zone
    - **Line 2c**: mineral zone data begins
    - **Line 2d**: *min _name* (string), *vol_frac* (real), *react_surf* (real)
        i. *min_name*: name of mineral in mineral zone (must be in the chemical system)
        ii. *vol_frac.:* initial volumetric fraction of mineral
        iii. *react_surf*: specific surface of mineral (surface of mineral per unit volume of rock)
    - **Line 2e**: end of mineral zone data
    - **Line 3**: end of section

```
‘INITIAL MINERAL ZONES’

1                                              ! number of mineral zones

1     25.0                                     ! index of mineral zone, temperature (C)

'calcite'                                      ! name of mineral zone

'mineral     vol.frac.   area'

'calcite'    0.50     1d2

'*'       0.00     0.00

'-------------------------------------------------------------------------------------------------------------------------------------------------'
```

- Section name: 'INITIAL AND BOUNDARY GAS ZONES'
  - **Line 1**: *num_gas_zones* (integer)
    i. Number of gas zones
  - **Line 2a**: *ind_gas_zone* (integer), *temp* (real), *vol* (real)
    i. *ind_gas_zone*: index of gas zone
    ii. *temp*: temperature of gas zone (in Celsius)
    iii. *vol*: total volume of gas
  - **Line 2b**: gas zone data begins
  - **Line 2c**: *name* (string)
    i. Name of gas zone
  - **Line 2d**: *gas _name* (string), *part_press* (real)
    i. *gas_name*: name of gas in gas zone (must be in the chemical system)
    ii. *part_press:* initial partial pressure (equivalent to activity) of gas (in atm)
  - **Line 2e**: end of gas zone data
  - **Line 3**: end of section
  - **Line 4**: end of file

```
'INITIAL AND BOUNDARY GAS ZONES'

1                                   ! number of gas zones

1    25    1d9                      ! index gas zone, temperature (C), volume

'atmosphere'                        ! name of gas zone

'gas              partial pressure'

'o2(g)'               2d-1

'co2(g)'              3.5d-4

'*'                   0.00

'-----------------------------------------------------------------------------------------------------------------------------------'

'end'
```

Waters file: *“_tar_wat.dat”*

This file contains the **association** between waters, target solids and/or target gases, and it also indicates if the waters are external or not. This file is read by the **Chemistry** class.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: ‘WATERS’
  - **Line 1**: *num_wat* (integer)
    - Number of waters
  - **Line 2**: *num_rech_wat* (integer)
    - Number of recharge waters
  - **Line 3**: *num_bd_wat* (integer)
    - Number of boundary waters
  - **Lines 4 - 4+nº intervals**: *int_wat* (string), *iwtype* (integer), *int_sol* (integer), *int_gas(*integer), *flag_wat_type*(logical)
    - *int_wat*: Interval of waters
    - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
    - *int_sol*: Interval of target solids associated with *int_wat*
    - *int_gas*: Interval of target gases associated with *int_wat*
    - *flag_wat_type:* Flag to indicate water type.
      - 0: Boundary
      - 1: Domain
      - 2: Recharge
  - End of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'--------------------------------------------------------------------------------------------------------------------------------------------------------'
'WATERS'
12                                     ! number of waters
0                                      ! number of recharge waters
2                                      ! number of boundary waters
1       1       1       1       0      ! waters interval, water type index, target solids interval, target gases interval, flag water type
2-11    2       1-10    0       1
12      2       10      0       0
'--------------------------------------------------------------------------------------------------------------------------------------------------------'
'end'
```

Target solids file: *"_tar_sol.dat"*

This file contains the **association** between target solids and solid zones. It is read by the **Chemistry** class. The only solid zones implemented so far are the **mineral** zones. **IMPORTANT**: this file will not be read by the code if there are no solid zones.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TARGET SOLIDS'
  - **Line 1**: *num_tar_sol* (integer)
    - Number of target solids
  - **Line 2**: *interval* (string), *iszone* (integer)
    - *interval*: Target solids interval
    - *iszone:* Index of mineral zone in section 'INITIAL MINERAL ZONES'
  - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-------------------------------------------------------------------------------------------------------'

'TARGET SOLIDS'

10                                      ! number of target solids

1-10      1                             ! target solids interval, solid zone index

'-------------------------------------------------------------------------------------------------------'

'end'
```

Target gases file: *"_tar_gas.dat"*

This file contains the **association** between target gases and gas zones. It is read by the **Chemistry** class. **IMPORTANT**: this file will not be read by the code if there are no gas zones.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TARGET GASES'
  - **Line 1**: *num_tar_gas* (integer)
    - Number of target gases
  - **Line 2**: *interval* (string), *igzone* (integer)
    - *interval*: Target solids interval
    - *igzone:* Index of gas zone in section 'INITIAL AND BOUNDARY GAS ZONES'
  - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-------------------------------------------------------------------------------------------------------'

'TARGET GASES'

1                                       ! number of target gases

1         1                             ! target gases interval, gas zone index

'-------------------------------------------------------------------------------------------------------'

'end'
```

Chemical output options file: *"_out_opts.dat"*

This file contains the **species, reactions, variables, time steps** and **targets** that the client wants to be **written** in the output file during a reactive mixing simulation. This file is read by the **Chemical Output Options** structure.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TIME STEPS'
  - **Line 1**: *nits* (integer)
    - Number of intermediate output time steps
  - **Line 2**: *time_steps* (integer vector)
    - Intermediate output time steps
  - end of section
- **Section name**: 'WATERS'
  - **Line 1**: *now* (integer)
    - Number of output waters
  - **Line 2**: *ind_wat* (integer vector)
    - Output waters indices
  - end of section
- **Section name**: 'TARGET SOLIDS'
  - **Line 1**: *nots* (integer)
    - Number of output target solids
  - **Line 2**: *ind_tar_sol* (integer vector)
    - Output target solids indices
  - end of section
- **Section name**: 'VARIABLES'
  - **Line 1**: *num_vars* (integer)
    - Number of output variables
  - **Lines 2-2+*num_vars*-1**: *name* (string)
    - *name*: Output variable name
    - **Important**: There are only **three** implemented variables so far:
      - *conc*: **concentration**
      - *react_rate*: **reaction rate**
      - *vol_fract*: **volumetric fraction**
  - end of section
- **Section name**: 'AQUEOUS SPECIES'
  - **Line 1**: *num_aq_species* (integer)
    - Number of output aqueous species
  - **Lines 2-2+*num_aq_species*-1**: *name* (string)
    - *name*: Aqueous species name
    - If *name*='all': All aqueous species will be considered
  - end of section
- **Section name**: 'MINERALS'
  - **Line 1**: *num_mins* (integer)
    - Number of output minerals
  - **Lines 2-2+*num_mins*-1**: *name* (string)
    - *name*: Mineral name
    - If *name*='all': All minerals will be considered
  - end of section

- **Section name**: 'REACTIONS'
  - **Line 1**: *num_reacts* (integer)
    - Number of output reactions
  - **Lines 2-2+*num_reacts*-1**: *name* (string)
    - *name*: Reaction name
    - If *name*='all': All reactions will be considered
  - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-------------------------------------------------------------------------------------------------------------------'
'TIME STEPS'
2                                   ! number of intermediate output time steps
1 5                                 ! intermediate time steps for output file (must be sorted in ascending order)
'-------------------------------------------------------------------------------------------------------------------'
'WATERS'
5                                   ! number of output waters
2 4 6 8 10                          ! waters for output file (must be sorted in ascending order)
'-------------------------------------------------------------------------------------------------------------------'
'TARGET SOLIDS'
3                                   ! number of output target solids
1 5 10                              ! target solids for output file (must be sorted in ascending order)
'-------------------------------------------------------------------------------------------------------------------'
'VARIABLES'                         ! names of variables that will be written in output file
3                                   ! number of variables
'conc'                              ! concentration
'react_rate'                        ! reaction rate
'vol_fract'                         ! volumetric fraction
-------------------------------------------------------------------------------------------------------------------'
'AQUEOUS SPECIES'                   ! names of aqueous species that will be considered in output file
5                                   ! number of aqueous species
'all'                               ! all aqueous species
-------------------------------------------------------------------------------------------------------------------'
'MINERALS'                          ! names of minerals that will be considered in output file
1                                   ! number of minerals
'all'                               ! all minerals
-------------------------------------------------------------------------------------------------------------------'
'REACTIONS'                         ! names of reactions that will be considered in output file
1                                   ! number of reactions
'calcite'                           ! calcite in equilibrium
-------------------------------------------------------------------------------------------------------------------'
'end'
```

Chemical computation options file: *"_comp_opts.dat"*

This file contains the different **computation options** for solving a reactive mixing problem with the WMA: the model to compute Jacobians, the WMA variant (consistent/lumped), the estimation method for reaction rates of downstream waters, and the way of averaging the implicit kinetic reaction rates. This file is read by the **Chemistry** class.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'COMPUTATION OPTIONS'
    - **Line 1**: *Jac_opt* (integer)
        - Option to compute Jacobians:
            - 0: incremental coefficients
            - 1: analytically
    - **Line 2**: *lump_flag* (logical)
        - .TRUE. if lumping is applied, .FALSE. otherwise
    - **Line 3**: *r_down_opt* (integer)
        - Option to estimate downstream waters reaction rates in consistent WMA
        - Four implemented options (see documentation for details):
            - 1: take values from previous time
            - 2: interpolate with values from two previous time steps by a linear combination
            - 3: interpolate using second derivative with values from three previous time steps
            - 4: assume that a fraction of the reaction takes place in the original target, and the remaining fraction in the current target
    - **Line 4**: *rk_avg_opt* (integer)
        - Option to average kinetic reaction rates in implicit WMA:
            - 1: first average concentrations and then compute reaction rates
            - 2: compute reaction rates and then average them
    - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'------------------------------------------------------------------------------------------------------------------------------------------------------------'

'COMPUTATION OPTIONS'

1                   ! Option to compute Jacobians (0: incremental coefficients, 1: analytically)

.true.              ! Lumping flag (.true. if lumped WMA, .false. if consistent WMA)

1                   ! Estimation of downstream waters reaction rates (4 options, see documentation for details)

2                   ! Averaging kinetic reaction rates option (1: concentrations, 2: reaction rates)

'-----------------------------------------------------------------------------------------------------------------------------------------------------------'

'end'
```

Convergence parameters file: *“_CV_params.dat”*

This file contains some **convergence parameters** for solving iterative methods, such as Newton-Raphson for speciation. These parameters are the absolute and relative tolerances, and the damping parameter for estimating reaction rates in downstream waters. The remaining parameters are set by default in the code, such as the maximum number of iterations allowed. This file is read by the **Convergence Parameters** structure.

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----------------------------------------------------------------------------------------------------------------------'

'CONVERGENCE PARAMETERS'

1d-14                          ! absolute tolerance

-14                            ! log absolute tolerance

1d-14                          ! relative tolerance

-14                            ! log relative tolerance

0d0                            ! Parameter for estimation of reaction rates

'-----------------------------------------------------------------------------------------------------------------------'
```